

FileHandling

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TEXT FILE

Example: **A**

↓

Character encoding

↓

01000001

8 bits = 1 byte

IMAGE FILE

Example: image data

↓

Image format encoding

↓

01010110 00101101 ...

bits grouped into bytes

STORAGE LEVEL

Both are ultimately stored as **bits (0s and 1s)**.

The difference is **how those bits are interpreted**: text as characters, image data according to an image format.

Bit

↓

0 or 1

↓

8 BITS

↓

One byte

↓

01000001 (Binary representation / Binary data)

(If we interpret these bits as a NUMBER)

↓

01000001 = 65

65 → DECIMAL NUMBER

(because we are representing it in base 10 (0 to 9))

If we look at 01000001 as a NUMBER SYSTEM

(because we are representing it in base 2)

↓

01000001 → BINARY NUMBER

(because it uses base 2: only 0 and 1)

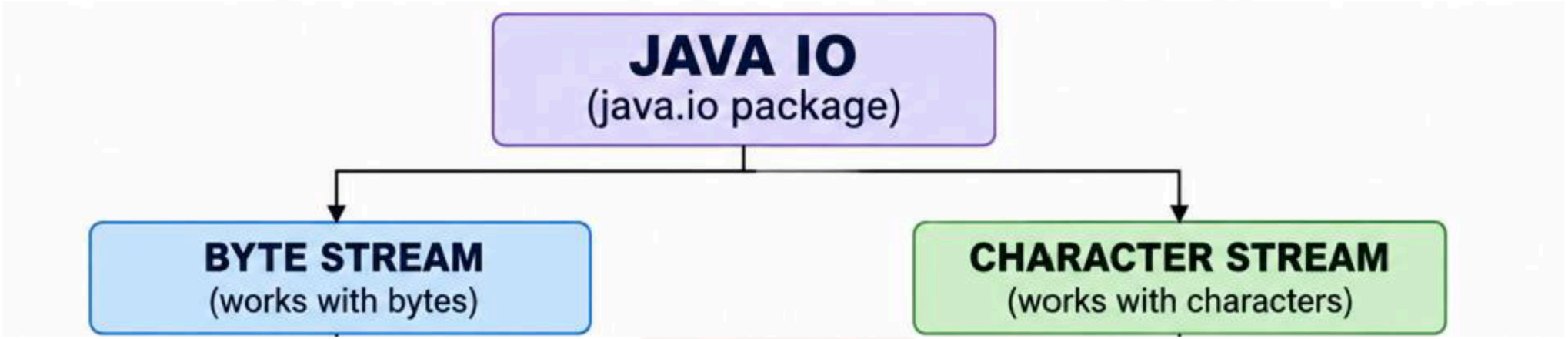
To read image file = Image viewer App
To read the text file = Notepad, Notepad++ , VS code etc..

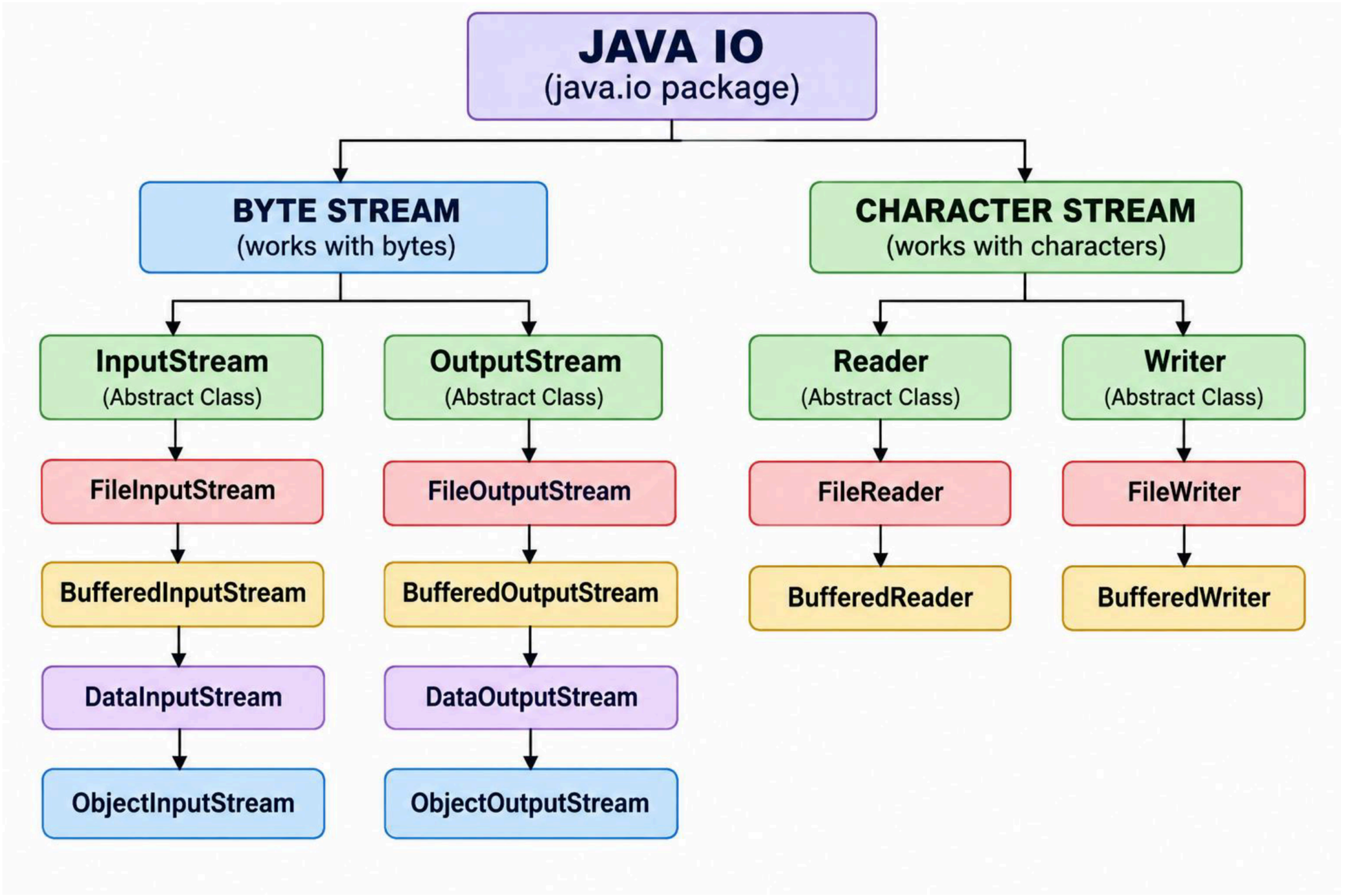
Greet.txt = Notepad++ -> Interprets Text -> Hello

java.jpeg = Image viewer -> Reads image -> Displays image

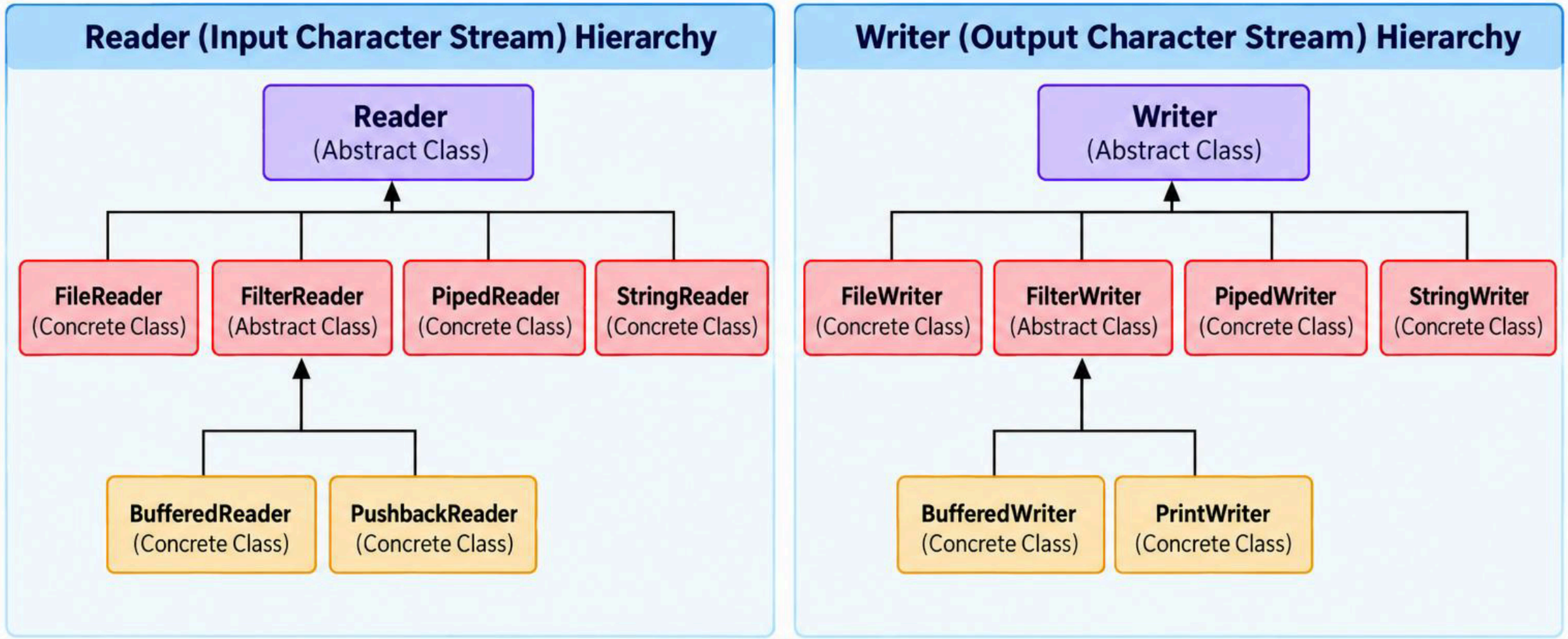
To interpret Raw bytes = Binary File
java.jpeg = Image viewer -> Reads image -> Displays image

To interpret text = Text File
Greet.txt = Notepad++ -> Interprets Text -> Hello


JAVA IO
(java.io package)
BYTE STREAM
(works with bytes)
CHARACTER STREAM
(works with characters)


JAVA IO
(java.io package)
BYTE STREAM
(works with bytes)
CHARACTER STREAM
(works with characters)
InputStream
(Abstract Class)
OutputStream
(Abstract Class)
Reader
(Abstract Class)
Writer
(Abstract Class)
FileInputStream
FileOutputStream
FileReader
FileWriter
BufferedInputStream
BufferedOutputStream
BufferedReader
BufferedWriter
DataInputStream
DataOutputStream
ObjectInputStream
ObjectOutputStream

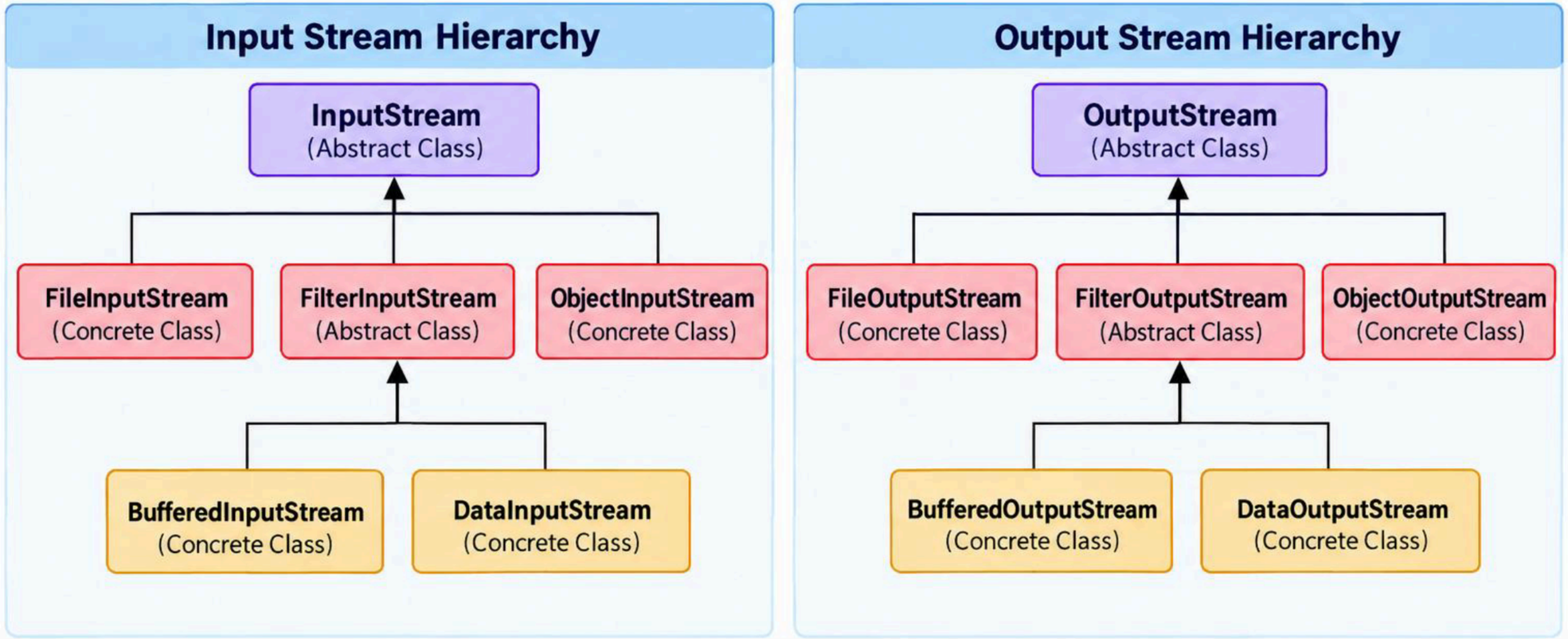
Java IO – Character Stream Hierarchy
(java.io package)



Legend:

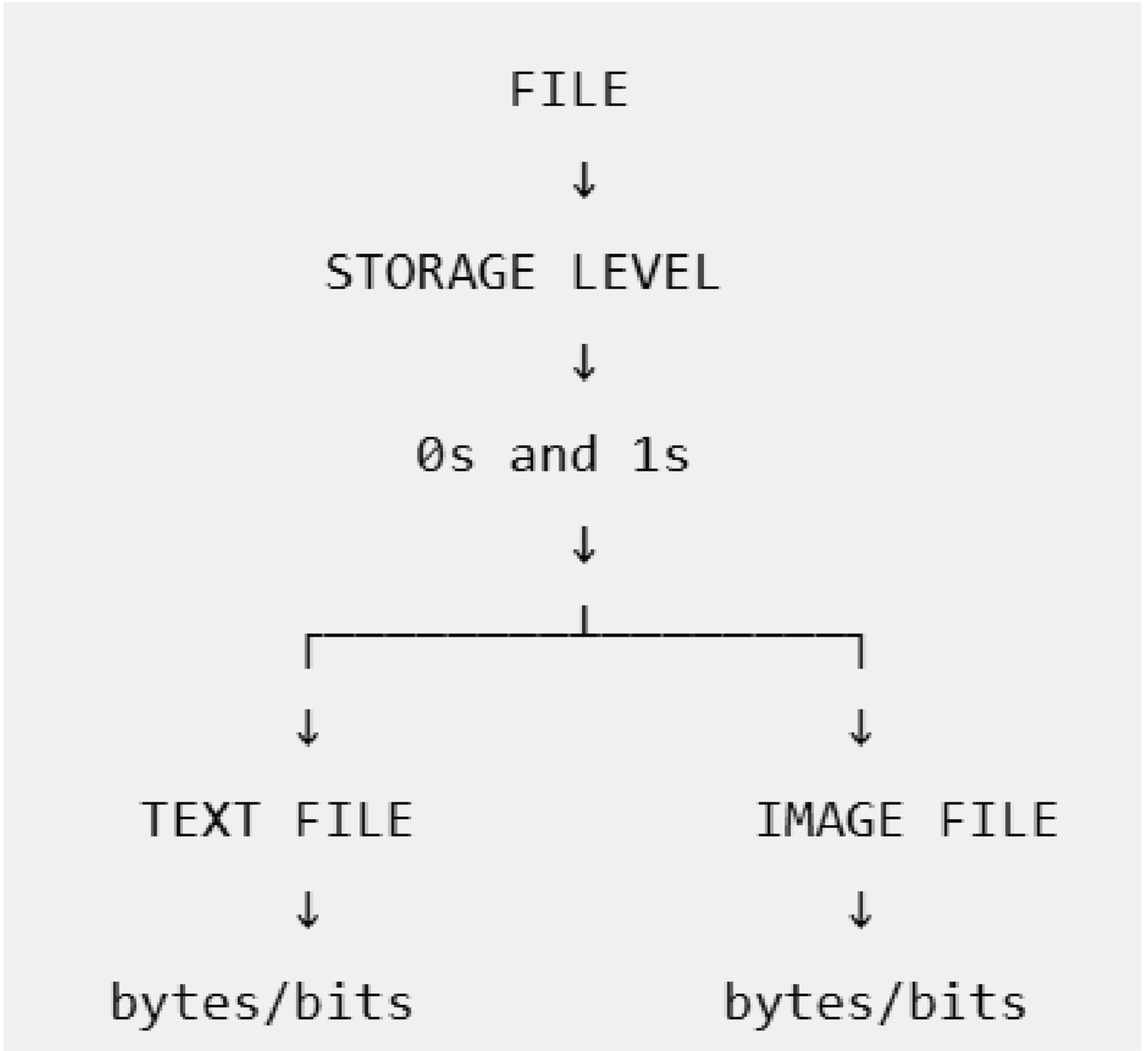
↑ extends
(points to super class)

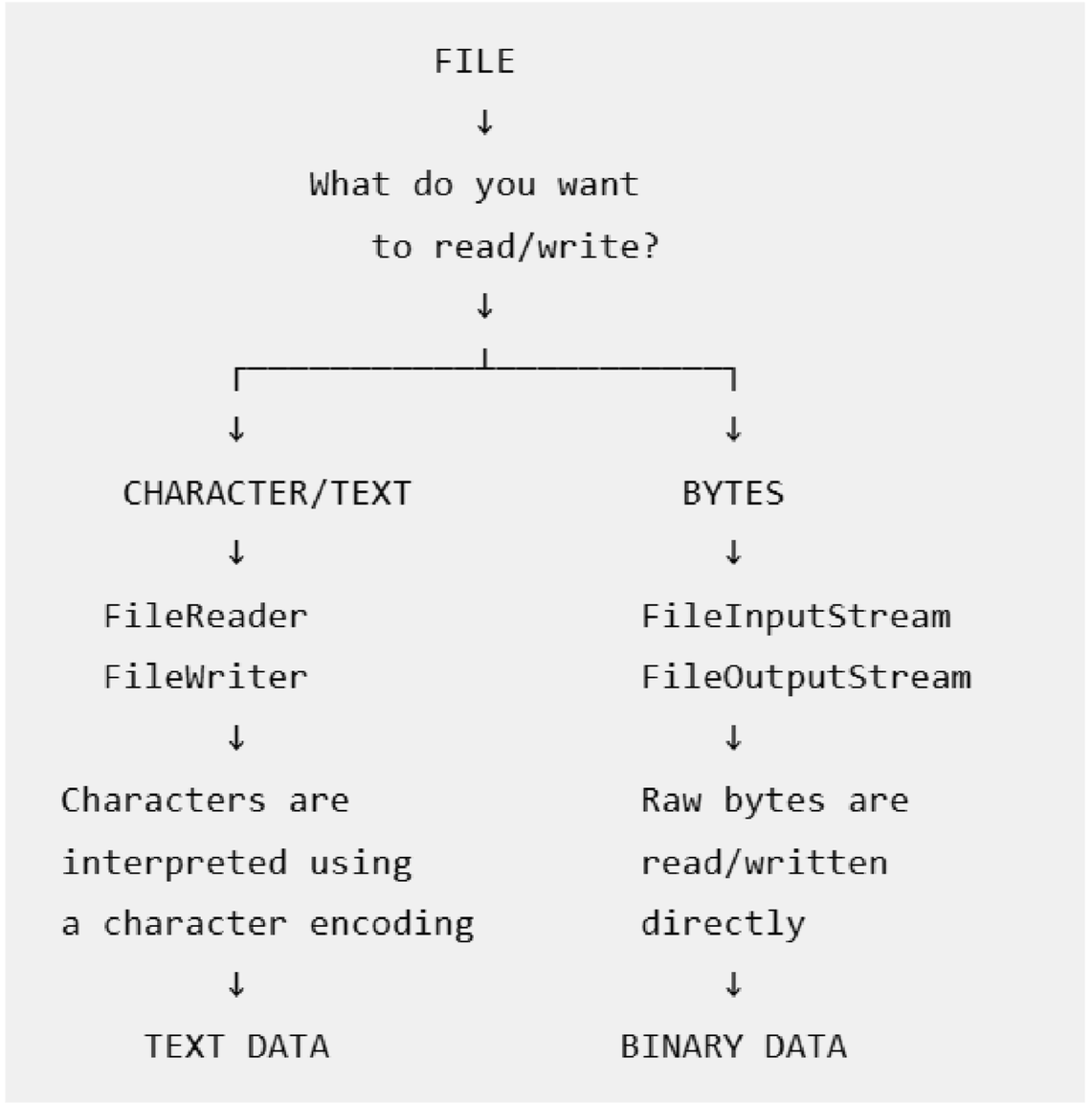
Java IO – Byte Stream Hierarchy
(java.io package)



Legend:

extends
(points to super class)


FILE
STORAGE LEVEL
0s and 1s
TEXT FILE
IMAGE FILE
bytes/bits
bytes/bits


FILE
What do you want
to read/write?
CHARACTER/TEXT
BYTES
FileReader
FileWriter
FileInputStream
FileOutputStream
Characters are
interpreted using
a character encoding
Raw bytes are
read/written
directly
TEXT DATA
BINARY DATA

Text File	Binary File
student.txt	photo.jpg
students.csv	song.mp3
data.xml	movie.mp4
config.properties	report.pdf
Human-readable text	Raw bytes
Character streams	Byte streams

read() reads one character at a time.

File → H → Java program
File → e → Java program
File → l → Java program
File → l → Java program
File → o → Java program
…

without buffering

100 books

house → 1 book → You
house → 1 book → You
house → 1 book → You
…
100 trips

50 book -> Car (buffer) -> reading

50 book -> Car (buffer) -> reading

File
↓
[ABCDE.....] ← one chunk
↓
Buffer
↓
Reads that chunk

File
↓
[FGHIJ.....] ← another chunk
↓
Buffer
↓
Reads that chunk

THANK YOU